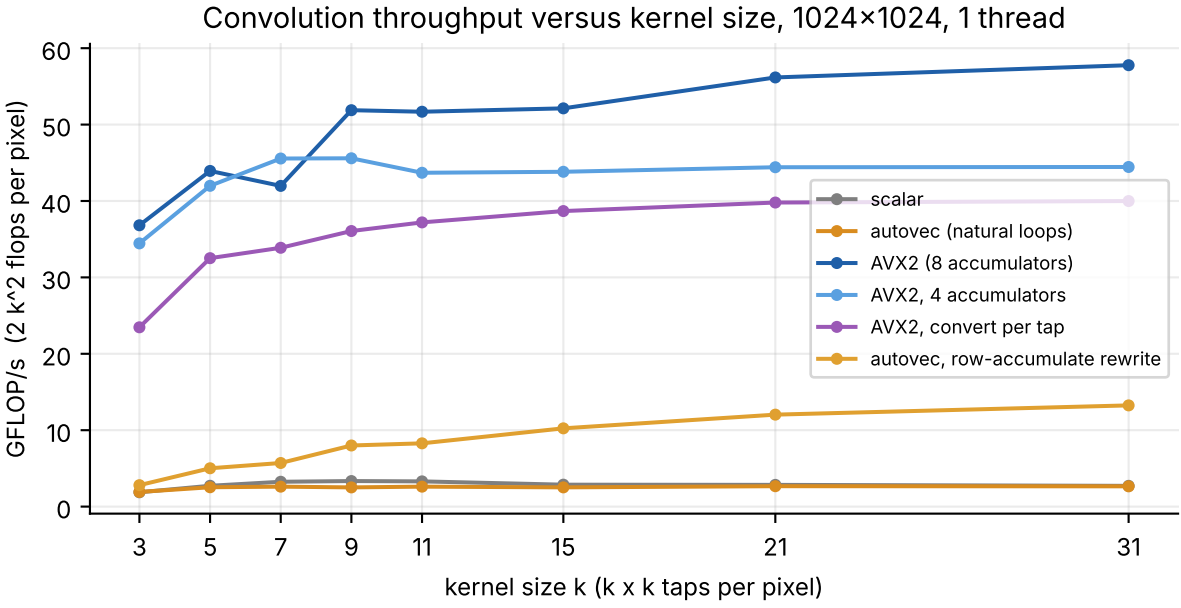
Convolution throughput versus kernel size, 1024×1024, 1 thread
GFLOP/s (2 k^2 flops per pixel)
60
50
40
30
20
10
0
3
5
7
9
11
15
21
31
kernel size k (k x k taps per pixel)
scalar
autovec (natural loops)
AVX2 (8 accumulators)
AVX2, 4 accumulators
AVX2, convert per tap
autovec, row-accumulate rewrite